

MLB Home Run Quality Analysis

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Tools: R, tidyverse, ggplot

Distinguishing True Power from Parks

Motivation

Context

Home Run totals don't separate true power from park effects. It also doesn't separate weak contact that carries in hitter-friendly environments.

Goal

I wanted to find a way to measure home run quality, not just quantity.

Why it Matters?

This provides more context for evaluating a player's underlying power across different park environments.

Research Questions

Question 1

How do MLB stadiums inflate or suppress home run outcomes relative to batted-ball quality?

Question 2

Which players generate the most—and least—park-independent home run production?

Question 3

How does adjusting for home run quality change player evaluation compared with traditional HR totals?

Data

Data was taken from Baseball Savant

Key Variables:

- Stadiums Gone In
- HR (1=Yes, 0=No)
- Hit Distance
- Exit Velocity
- Each batted ball included was a homerun in at least one park

Date	Batter	Pitcher	🏏	B	🏏	TB	🏏	🏏	C	🏏	K	🏏	🏏	A	A	🏏	T	🏏	🏏	P	🏏	🏏	C	🏏	P	🏏	A	🏏	🏏	🏏	🏏	Gone At	Result	
2026-05-10	 A. Judge	 L. Henderson	✓	✗	✓	✓	✓	✓	✓	✗	✓	✗	✗	✓	✗	✗	✓	✓	✗	✗	✓	✓	✓	✓	✗	✗	✓	✓	✓	✗	✗	16	Home Run	
2026-05-06	 A. Judge	 N. Eovaldi	✓	✗	✓	✓	✓	✓	✓	✓	✗	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	27	Home Run	
2026-05-04	 A. Judge	 S. Baz	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	28	Home Run	
2026-05-03	 A. Judge	 T. Gibson	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	30	Home Run	
2026-04-28	 A. Judge	 C. Winn	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	30	Home Run	
2026-04-27	 A. Judge	 J. Leiter	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	30	Home Run	
2026-04-26	 A. Judge	 S. Arrighetti	✓	✗	✗	✗	✓	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	✓	✗	✓	✓	24	Home Run	

Home Run Classification System

Home Run in
1-7 Stadium

Doubter

Home Run in
8-23 Stadiums

Mostly Gone

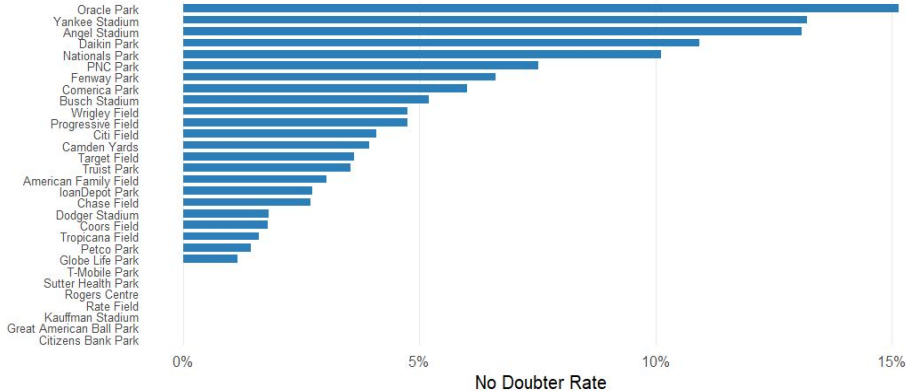
Home Run in
24-30 Stadiums

No Doubter

Stadium Effects on Home Run Quality

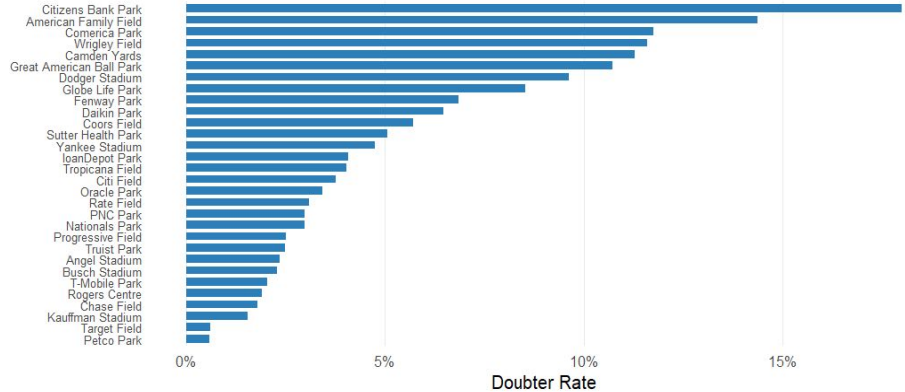
Non-Home Run No Doubter Rate by Stadium

Share of home runs that would leave in 24 or more parks but don't



HR Doubter Rate by Stadium

Share of home runs that would leave 7 or fewer MLB parks



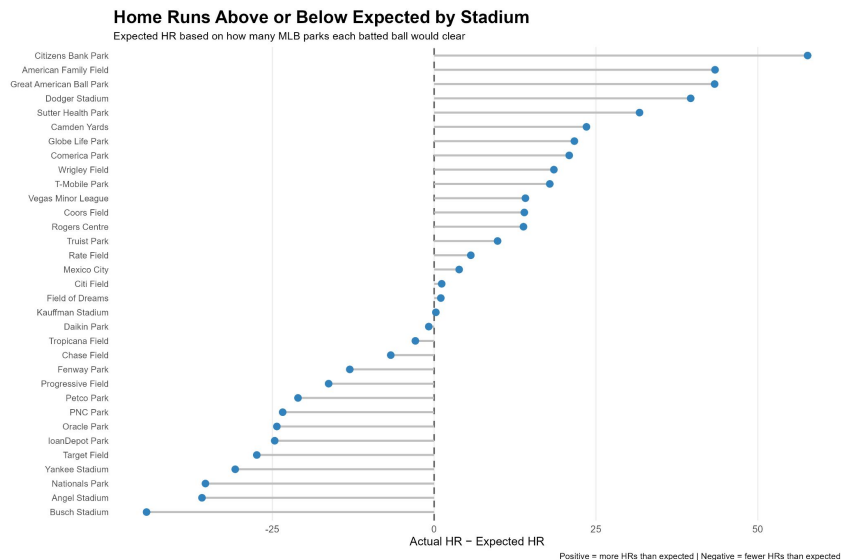
Key Findings

- **Oracle Park has the highest rate of high-quality contact staying in the park, at roughly 15%.**
- Yankee Stadium and Angel Stadium also show relatively high rates of batted balls that would clear **24+ parks but did not become HRs.**
- This provides evidence of the opposite park effect: some stadiums may **suppress otherwise HR-quality contact.**

Key Findings

- **Citizens Bank Park stands out, with roughly 18% of its HRs classified as Doubters.**
- Several parks show much higher rates of **park-dependent HRs** than others.
- This suggests park dimensions can help turn borderline batted balls into home runs.

Ballpark Effects on Home run Outcomes



Key Findings

Citizens Bank Park showed the largest positive difference, producing roughly **57 more HRs than expected**.

Busch Stadium showed the largest negative difference, with roughly **45 fewer HRs than expected**.

The wide range across parks suggests **ballpark dimensions meaningfully influence whether similar batted balls become home runs**.

Methodology:

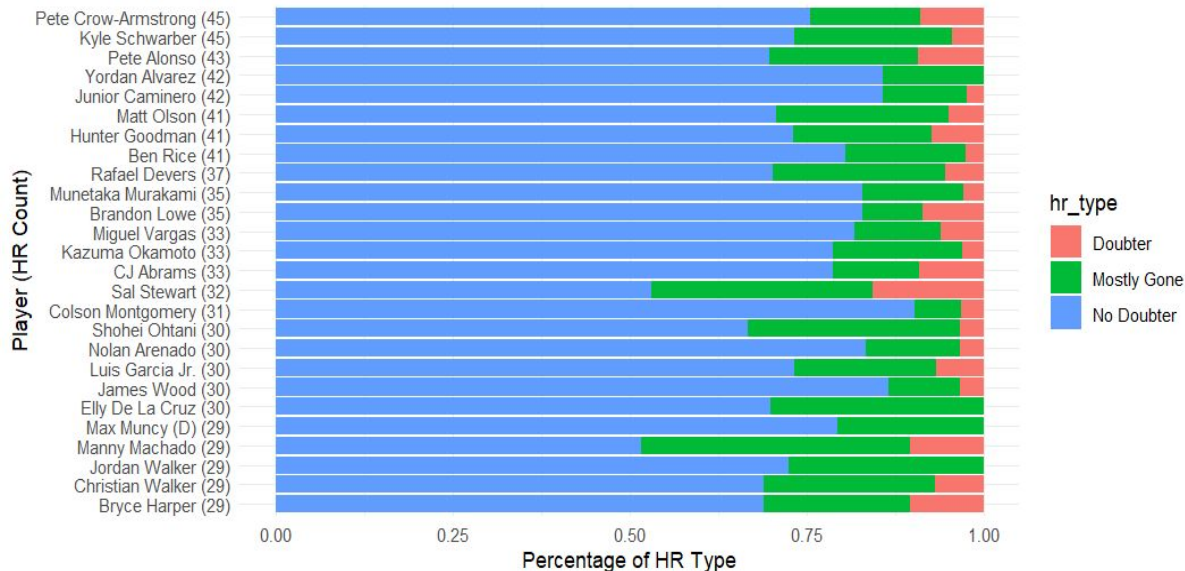
$xHR = \text{Parks Out} / 30$

$\text{Expected HR} = \sum xHR$

$\text{Park Difference} = \text{Actual HR} - \text{Expected HR}$

Player Profiles

HR Profile by Player



Key Findings

- **Yordan Alvarez and Junior Caminero stand out for pure HR quality**, with very high No Doubter shares while still producing 40+ HRs.
- **Sal Stewart appears to have one of the most park-dependent profiles** on the chart, with a much smaller No Doubter share and substantially more Mostly Gone/Doubter HRs. Same with **Manny Machado** receiving a similar type of park dependency.

Home Run Quality Score Model

1. Classify each HR

- No Doubter = 3 pts
- Mostly Gone = 2 pts
- Doubter = 1 pt

2. Calculate average HR quality

Average quality score
across all HRs

3. Adjust HR volume

Quality-Adjusted HR =
HR Total × Quality
Multiplier

$$\text{quality_multiplier} = 0.75 + (\text{avg_hr_quality} / 3) * 0.25$$

Quality adjusts the HR total rather than replacing it, preventing a small number of high-quality HRs from outweighing overall production.



Home Run Quality Adjustment and Rank

How HR Quality Changes Home Run Production

Top 20 players by actual home runs



How HR Quality Changes Home Run Production

Top 20 players ranked by actual home run totals

Player	Home Run Totals		Advanced Quality Metrics			
	Actual	Quality-Adj.	+ / -	Quality Score (1-3)	No-Doubter %	Avg Exit Velo (mph)
Kyle Schwarber	45	43.8	-1.2	2.7	73.3%	106.3
Pete Crow-Armstrong	45	43.8	-1.2	2.7	75.6%	104.1
Pete Alonso	43	41.6	-1.4	2.6	69.8%	106.4
Junior Caminero	42	41.4	-0.6	2.8	85.7%	106.9
Yordan Alvarez	42	41.5	-0.5	2.9	85.7%	107.0
Ben Rice	41	40.2	-0.8	2.8	80.5%	103.9
Hunter Goodman	41	39.8	-1.2	2.7	73.2%	105.1
Matt Olson	41	39.8	-1.2	2.7	70.7%	106.1
Rafael Devers	37	35.9	-1.1	2.6	70.3%	104.0
Brandon Lowe	35	34.2	-0.8	2.7	82.9%	103.9
Munetaka Murakami	35	34.4	-0.6	2.8	82.9%	106.8
CJ Abrams	33	32.2	-0.8	2.7	78.8%	103.0
Kazuma Okamoto	33	32.3	-0.7	2.8	78.8%	106.5
Miguel Vargas	33	32.3	-0.7	2.8	81.8%	104.1
Sal Stewart	32	30.3	-1.7	2.4	53.1%	104.3
Colson Montgomery	31	30.7	-0.3	2.9	90.3%	105.3
Elly De La Cruz	30	29.2	-0.8	2.7	70.0%	106.9
James Wood	30	29.6	-0.4	2.8	86.7%	108.6
Luis Garcia Jr.	30	29.2	-0.8	2.7	73.3%	105.4
Nolan Arenado	30	29.5	-0.5	2.8	83.3%	101.6

Sal Stewart takes one of the largest hits from the adjustment, driven by his unusually low No Doubter rate (~53%), suggesting his raw HR total may overstate his underlying power.

Colson Montgomery moves in the opposite direction, with about **90% of his HRs classified as No Doubters**, causing him to gain ground relative to players with similar HR totals.

Home Run Evaluation Insights

1st Insight

Stadiums can both inflate and suppress home run totals. Some parks produced more HRs than expected based on batted-ball quality, while others kept high-quality potential HRs in play.

2nd Insight

Players with similar HR totals can have very different power profiles. Some consistently generate No Doubters, while others rely more heavily on park-dependent HRs.

3rd Insight

Adjusting HR totals for quality creates separation that raw HR totals miss, rewarding players who consistently generate more portable home runs.

Why looking Past Home Run Total Matters

- Traditional home run totals can overstate or understate true power ability.
- Incorporating home run quality provides a more stable signal of underlying offensive skill.
- This helps better separate sustainable power hitters from players benefiting from context (parks, weak contact HRs, variance).

